

Systems, Processes, and Technology for Accounting

Spring 2026

Professor: Johan Perols

Email: jperols@ucsd.edu

Class Time: Wednesdays 2:00 pm-4:50 pm.

Course Delivery: The course will be delivered using live in-person lectures, recorded content, and written instructions. Recorded content and written instructions are accessible through Canvas. Class time will be used for lectures, assessments (skills tests and exams), and independent work on assignments with individual and small group help. I will not have scheduled office hours (outside of class times), but I am available for help throughout the week (including on weekends). For additional help, [Piazza](#) is the quickest and most effective way to communicate with me and the course TA.

Teaching Assistant: Karthick Raja Suriyanarayanamy (ksuriyanarayanamy@ucsd.edu)

DESCRIPTION

Develop an understanding of transaction cycle processing steps and data. Gain hands-on experience processing transactions in a manual accounting information system and analyzing transaction data using Microsoft Excel, SQL, and Tableau.

OBJECTIVES

At the close of Systems, Processes, and Technology for Accounting you will be able to:

- Use manual accounting information systems to process transactions, including completing various source and destination documents, recording transactions in journals, and posting to ledgers.
- Describe basic relational database concepts.
- Describe basic principles of visual perception and design thinking related to data visualization.
- Identify appropriate visualizations for different data types and best practices for creating visualizations.
- Analyze transaction data using Microsoft Excel, SQL, and Tableau:

Excel

- Nested if statements, and/or logic, vlookup, average, averageif, sum, sumif, stdev, min, max, count, countif, left, right, len, find, conditional formatting, date, datevalue, year, month, pivot tables, and simple macros.

SQL

- Create SELECT queries with: from, where (with =, is, >, <, in, between, like, and/or), order by, as, top, round, concatenate, substring, instr, trim, length, soundex, isnull, and nvl.
- Create aggregate queries using group by (with avg, count, max, min, and sum aggregate functions) and having and where.
- Use joins (inner, outer, and cross joins) to combine multiple tables
- Create subqueries and views.

Tableau

- Join tables using both inner and outer joins, calculate fields using IF logic, string and date expressions, and formulas, and correctly classify fields as dimensions or measures.
- Perform interactive data visualization to perform exploratory and confirmatory data analysis of transaction data including understanding pills, marks, different charts and chart formatting, filters, aggregations, control charts, table calculations, ranking, top, LOD expressions, etc.

MATERIALS

- PowerPoint slides, study guides, and other downloads will be posted on the course website. This content is essential and is the only thing together with the lectures that you need for exams.
- Systems Understanding Aid, 10th Edition, Arens and Ward (9780912503752).

GRADE COMPONENTS

Quizzes

There are four quizzes. Each quiz contains multiple choice questions. Any topic assigned for class or presented is possible quiz material. No collaboration allowed (I). The quizzes do not require calculators. *If you miss a quiz without prior permission, you will automatically receive an "F" grade for the exam.*

Cases/Assignments/Skills Assessments

Systems Understanding Aid (SUA) - Transaction processing systems are used to record and process transactions such as sales orders and purchase orders. The SUA case is a hands-on exercise that provides in-depth understanding of records (journals and ledgers), documents, and processes steps in an AIS. While collaboration in the form of discussion of the cases is allowed, each individual should complete his/her own work and copying each other is not allowed (I).

Analyzing Transaction Data

You will learn to analyze transaction data using three tools, Excel, SQL, and Tableau. Each of these tools will be introduced using an ungraded take-home assignment followed by a graded skills test (a shorter version of the take-home assignment containing similar problems to the more difficult exercises from the take-home assignment).

Excel - Excel is extensively used by business professionals and employers more or less expect that you know how to use Excel. This assignment will introduce you to Excel and also give you hands on experience with some of the more advanced data analysis functions in Excel. Some of the advanced features you will learn include vlookups, pivot tables, string and date functions, sumif and sumifs, and nested if statements. The Excel project consists of an ungraded take-home assignment (C) and a shorter in-class assignment that is graded (I). You do not need to submit the take-home assignment.

SQL - A large majority of business operational data and almost all accounting data are stored in traditional data systems, such as relational databases. The data in these databases are accessed and manipulated using SQL. Consequently, SQL skills is one of the most sought after skill by recruiters that hire business analytics professionals (Mamonov et al. 2014; Wixom et al. 2014). While SQL is widely used, many accountant are either not aware of the benefits SQL can provide or they lack the skills necessary to use SQL. You will be introduced to SQL and gain hands on experience using SQL to analyze transactional data. You will learn a wide variety of SQL features such as joins, find unmatched queries, case statements, complex where and having clauses, aggregate functions, views, and sub-queries. The SQL project consists of an ungraded take-home assignment (C) and a shorter in-class assignment that is graded (I). You do not need to submit the take-home assignment.

Tableau - Interactive data visualization (e.g., Tableau) is increasingly used to also explore and analyze structured data. Similarly to SQL, interactive data visualization uses filters, joins, and aggregations, but instead of producing tabular results, results are displayed as visualizations (e.g., diagrams). Interactive data visualization has quickly gained in importance (Pacampara 2014) and such skills are increasingly required by employers looking to hire data analytics professionals (Mamonov et al. 2014). Tableau and other interactive data visualization tools are also increasingly used in both industry and public accounting. You will learn a number of graph features in Tableau including creating different types of graphs, using the marks area, using trend lines, using quick table calculations, and creating calculated fields using if logic and LOD statements. The Tableau project consists of an ungraded take-home assignment (C) and a shorter in-class assignment that is graded (I). You do not need to submit the take-home assignment.

Point Allocation

Grade Component	Points
Quizzes (4 @ 5 pts)	20 points
Systems Understanding Aid	20 points
Excel Skills Test	20 points
SQL Skills Test	20 points
Tableau Skills Test	20 points
Total	100 points

GRADES

I do not round up, the points you earn determine your course grade

Course Point Percentage	Grade
93% or better	A
90% to 93%	A-
87% to 90%	B+
83% to 87%	B
80% to 83%	B-
77% to 80%	C+
73% to 77%	C
70% to 73%	C-
60% to 70%	D
0% to 60%	F

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind. The complete UCSD Policy on Integrity of Scholarship can be viewed at: <http://senate.ucsd.edu/Operating-Procedures/Senate-Manual/Appendices/2>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or osd@ucsd.edu.

TENTATIVE SCHEDULE

Dates	Topic	Due before class
1-Apr	REMOTE - Course introduction - SUA revenue transactions	
8-Apr	Quiz: sales processing lectures - SUA revenue transactions	
15-Apr	Quiz: sales controls lectures - SUA revenue/expenditure transactions	SUA revenue transactions (nothing to turn in)
22-Apr	Quiz: purchase processing lectures - SUA expenditure transactions	
29-Apr	Quiz: purchase controls lectures DUE: SUA - Excel take-home assignment	SUA
6-May	Excel take-home assignment	
13-May	Skills test: Excel - SQL take-home assignment	
20-May	SQL take-home assignment	
27-May	Skills test: SQL - Tableau take-home assignment	
3-June	Tableau take-home assignment	
10-June	5:00am-6:00pm Skills test: Tableau	

Bolded items are graded